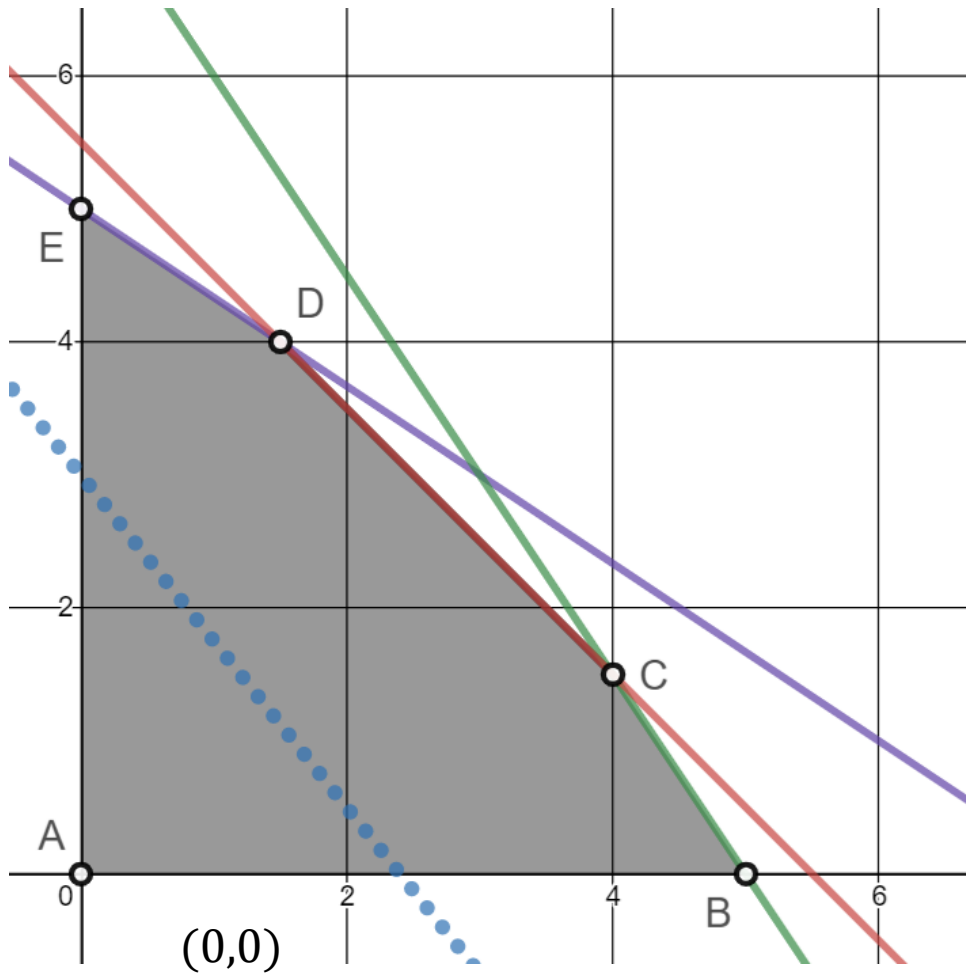


Simplex Method Graphical Demo



maximise

$$P = 5x + 4y$$

subject to:

$$2x + 3y \leq 15$$

$$3x + 2y \leq 15$$

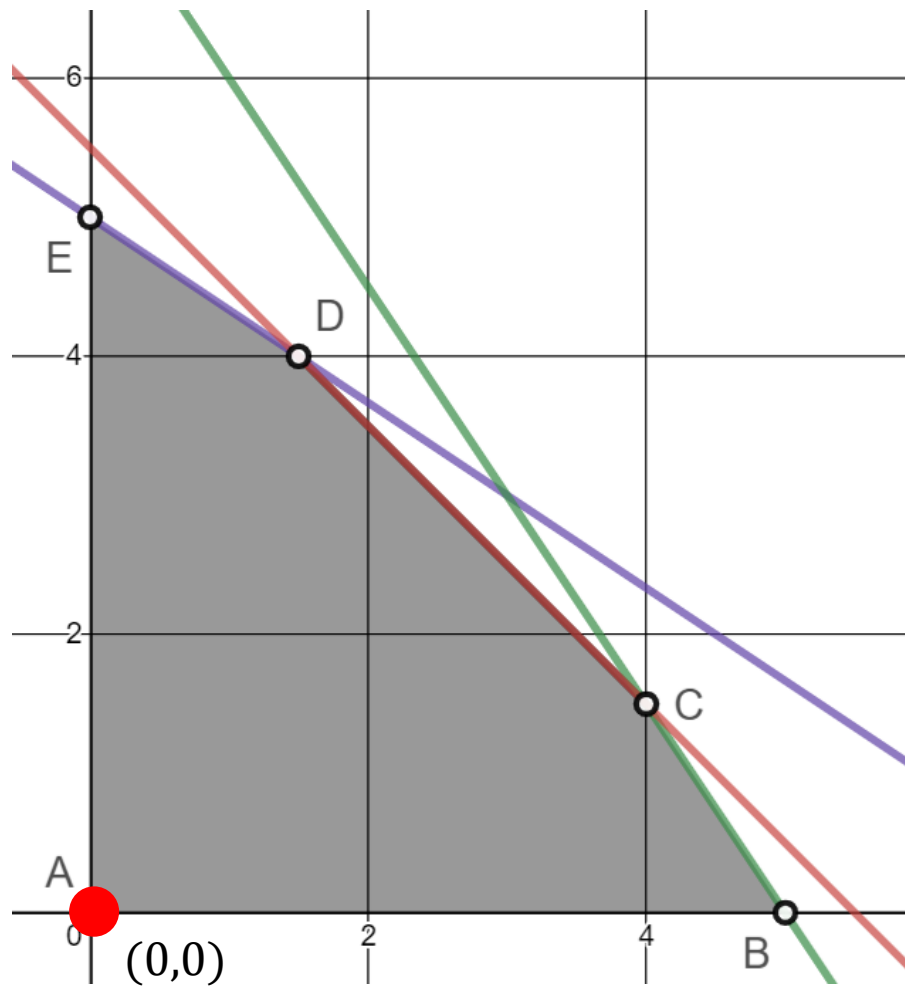
$$2x + 2y \leq 11$$

and

$$x, y \geq 0$$

Point A has an objective value of

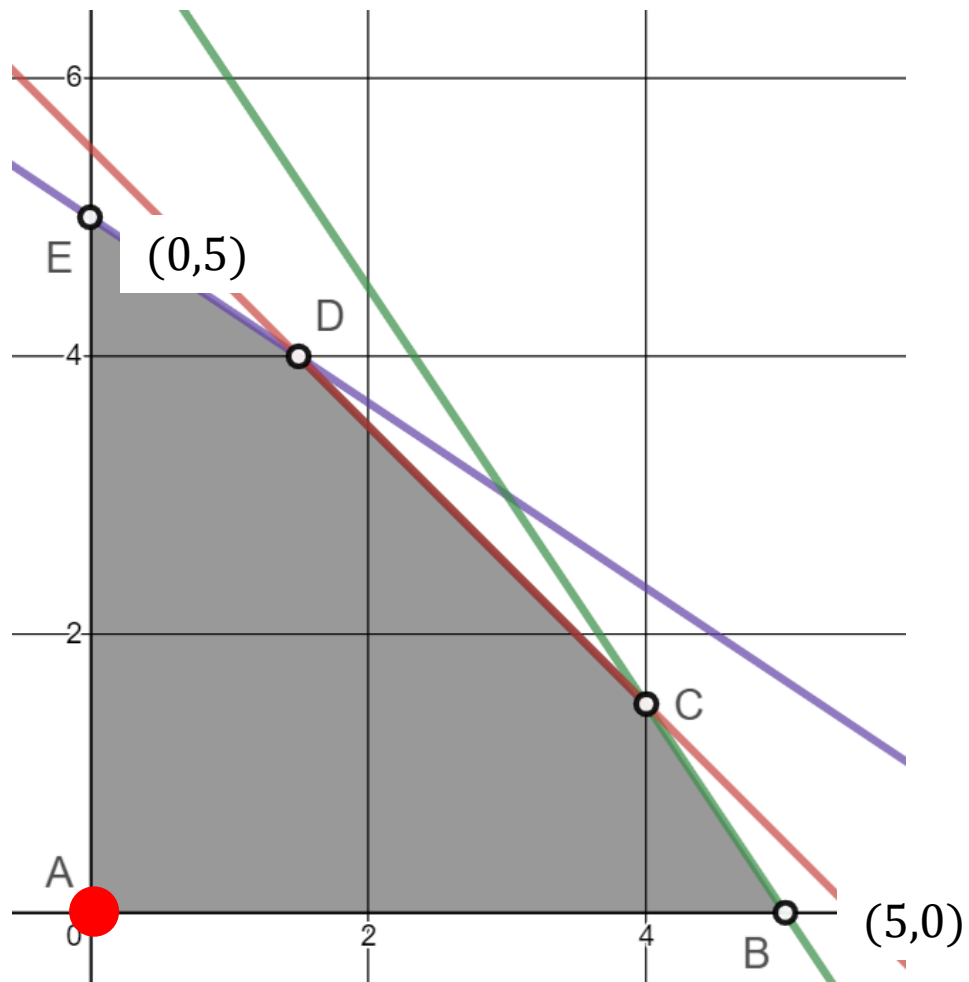
$$P = 5 \cdot 0 + 4 \cdot 0 = 0$$



$$P = 5x + 4y$$

Current Point	Adjacent Points	
A = 0		

Point A has an objective value of
 $P = 5 \cdot 0 + 4 \cdot 0 = 0$

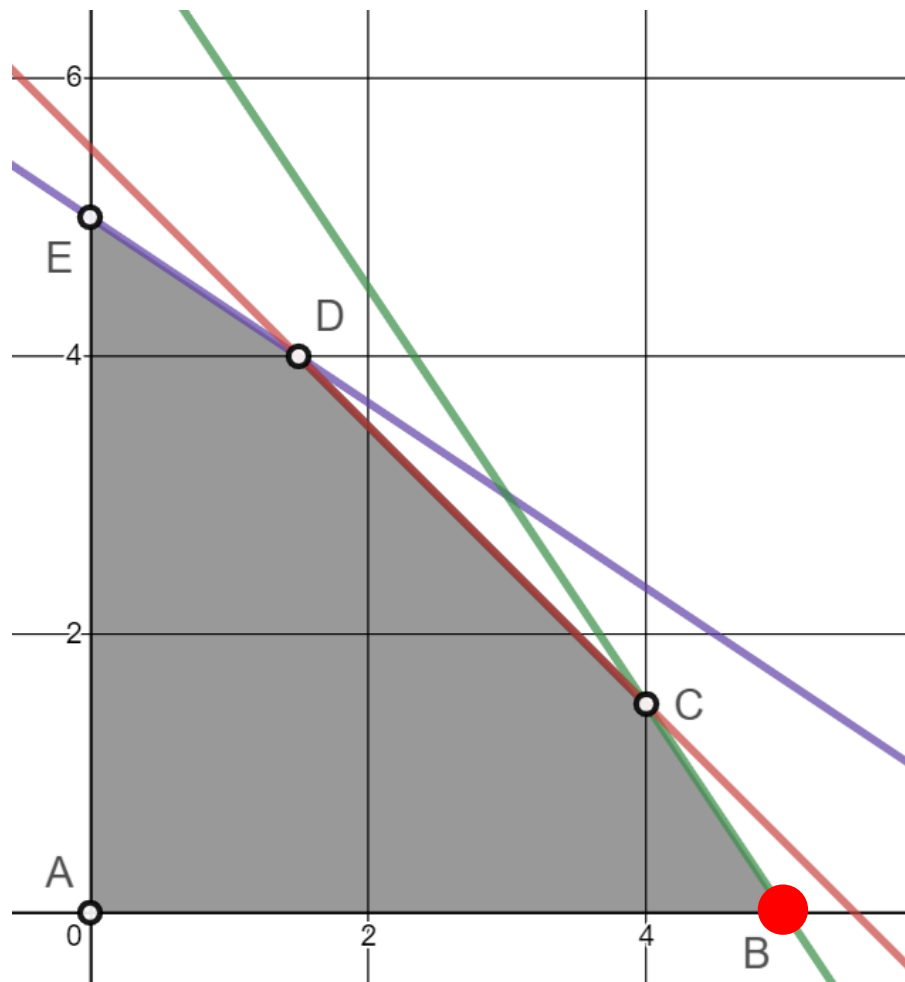


$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$

Point B (5,0) has an objective value of
 $P = 5 \cdot 5 + 4 \cdot 0 = 25$

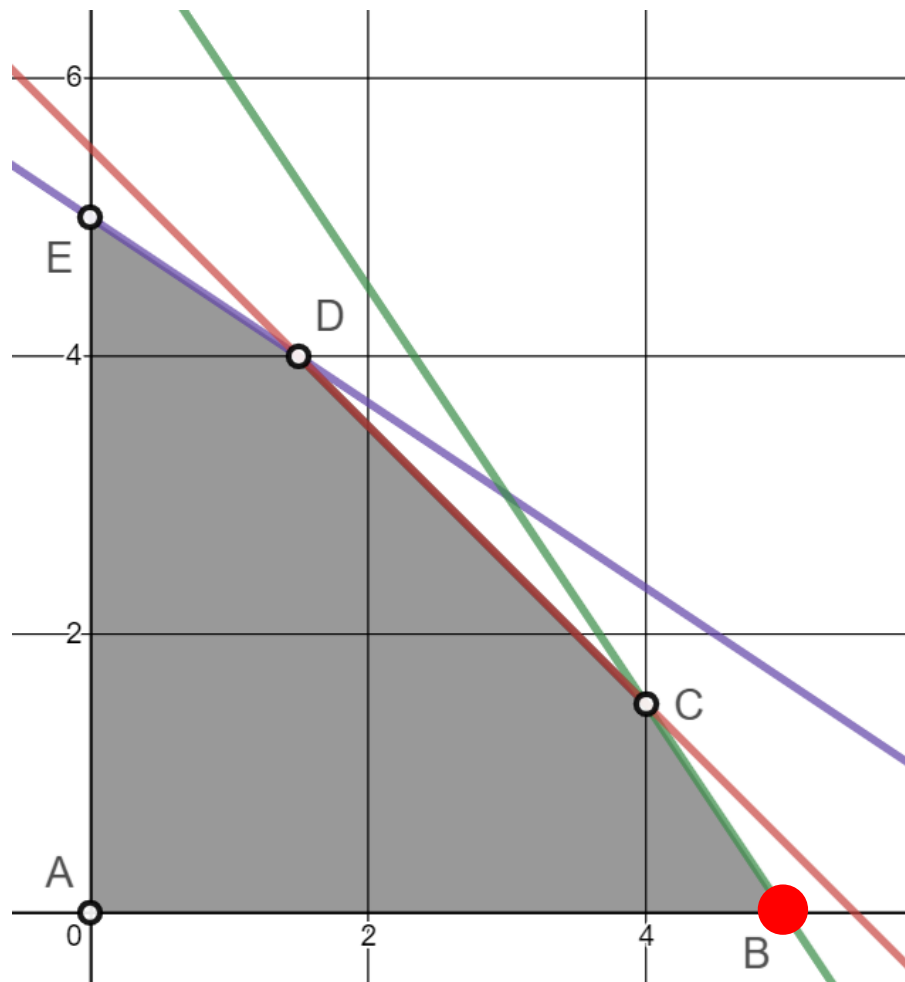
Point E (0,5) has an objective value of
 $P = 5 \cdot 0 + 4 \cdot 5 = 20$



$$P = 5x + 4y$$

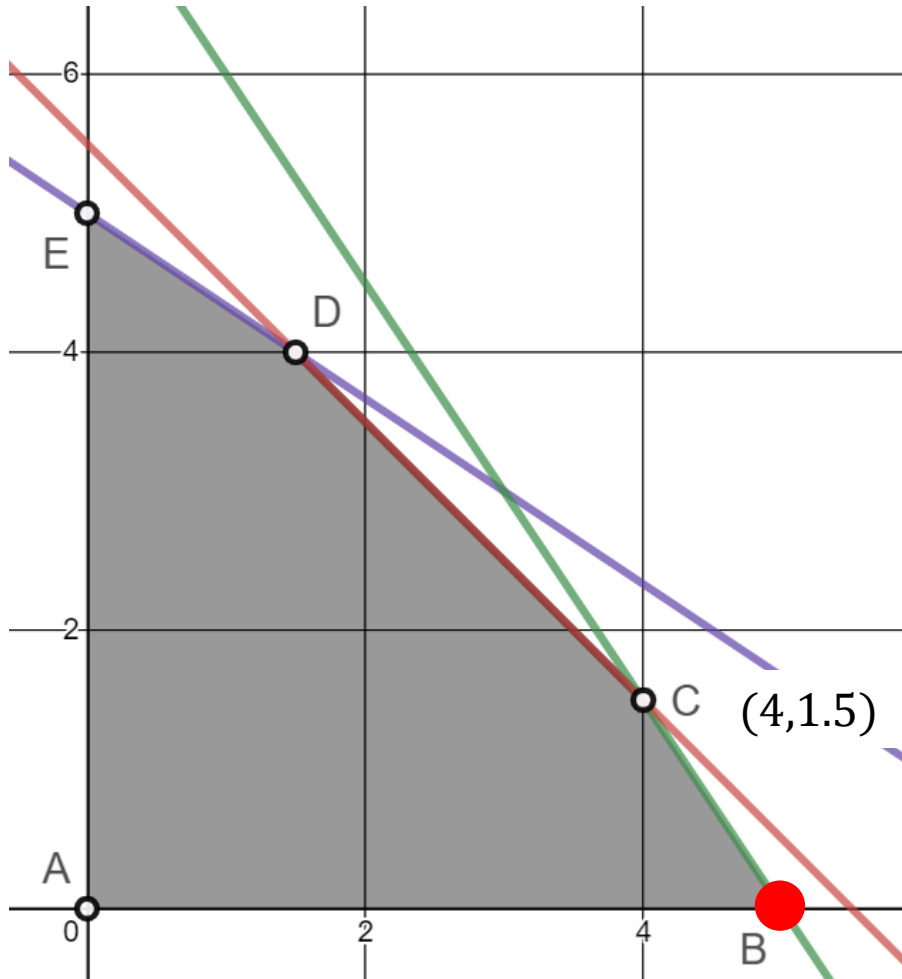
Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$

Choose B to move to!



$$P = 5x + 4y$$

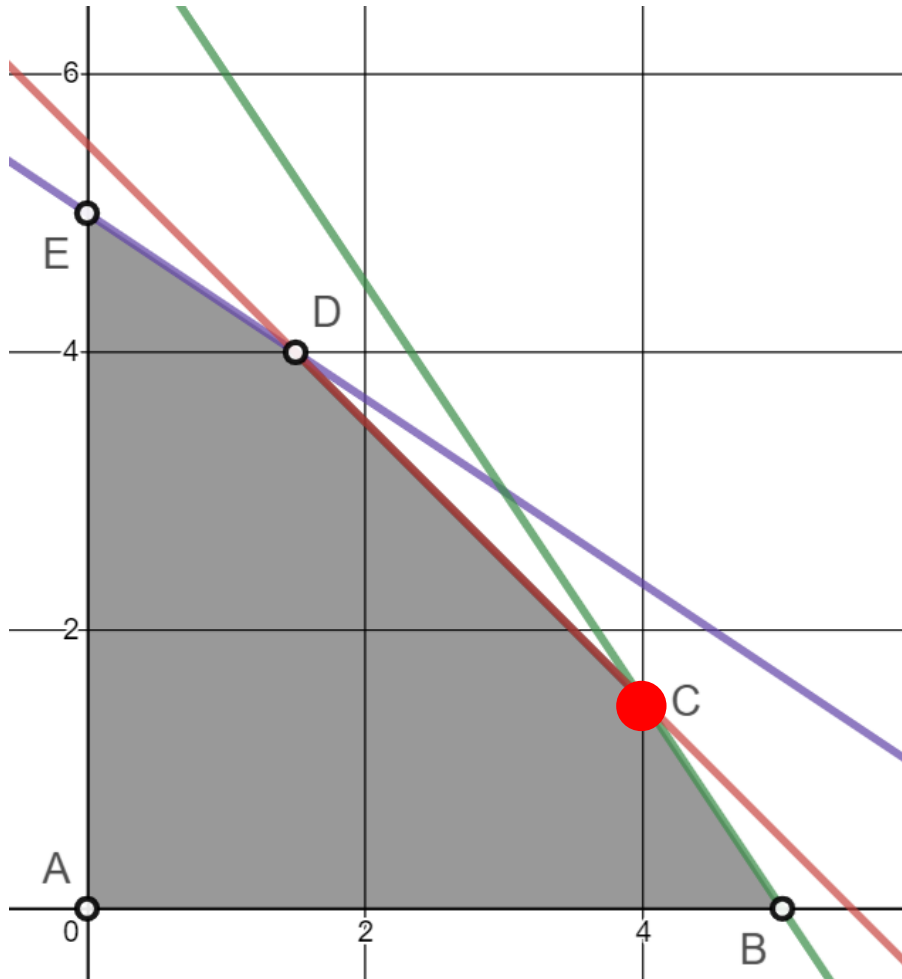
Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$		



$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$

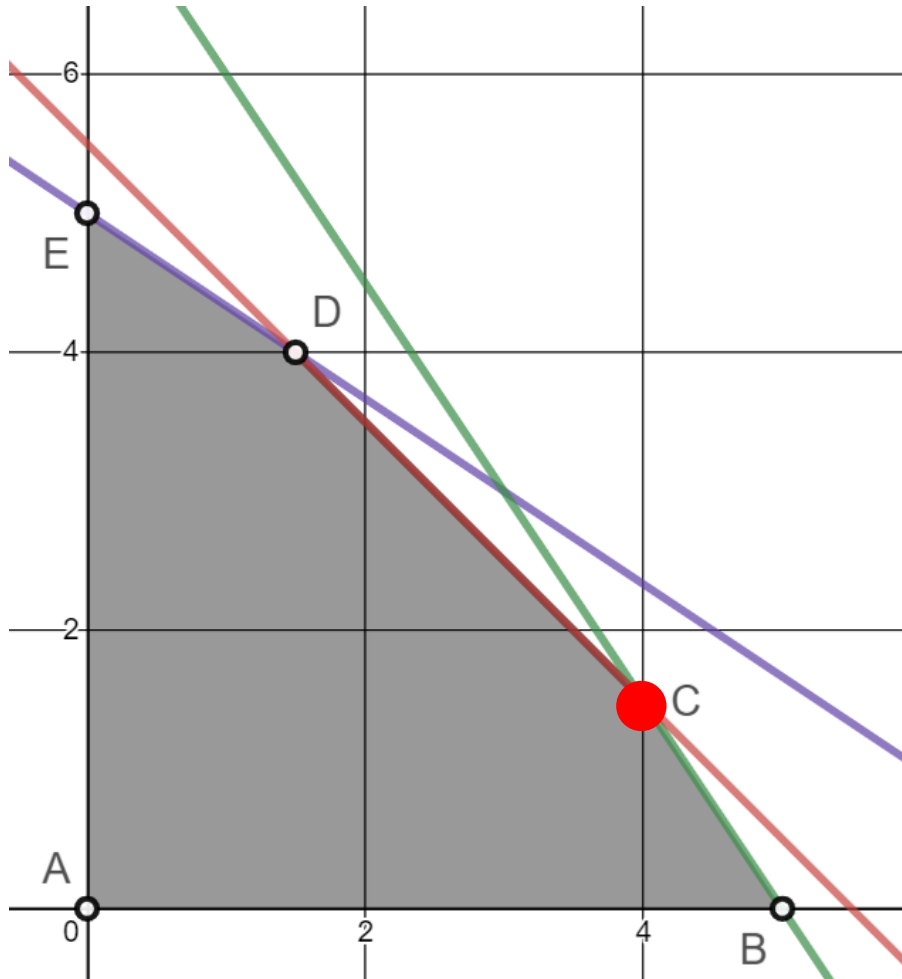
Point C (4,1.5) has an objective value of
 $P = 5 \cdot 4 + 4 \cdot 1.5 = 26$



$$P = 5x + 4y$$

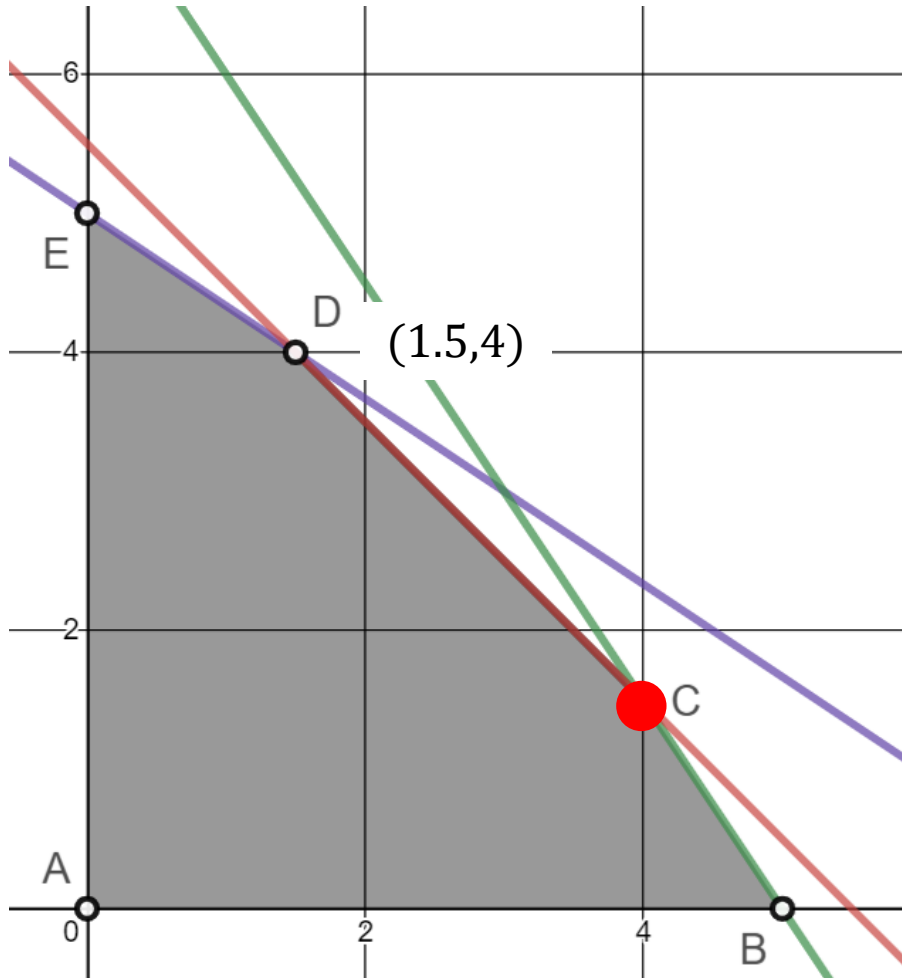
Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$

Choose C to move to!



$$P = 5x + 4y$$

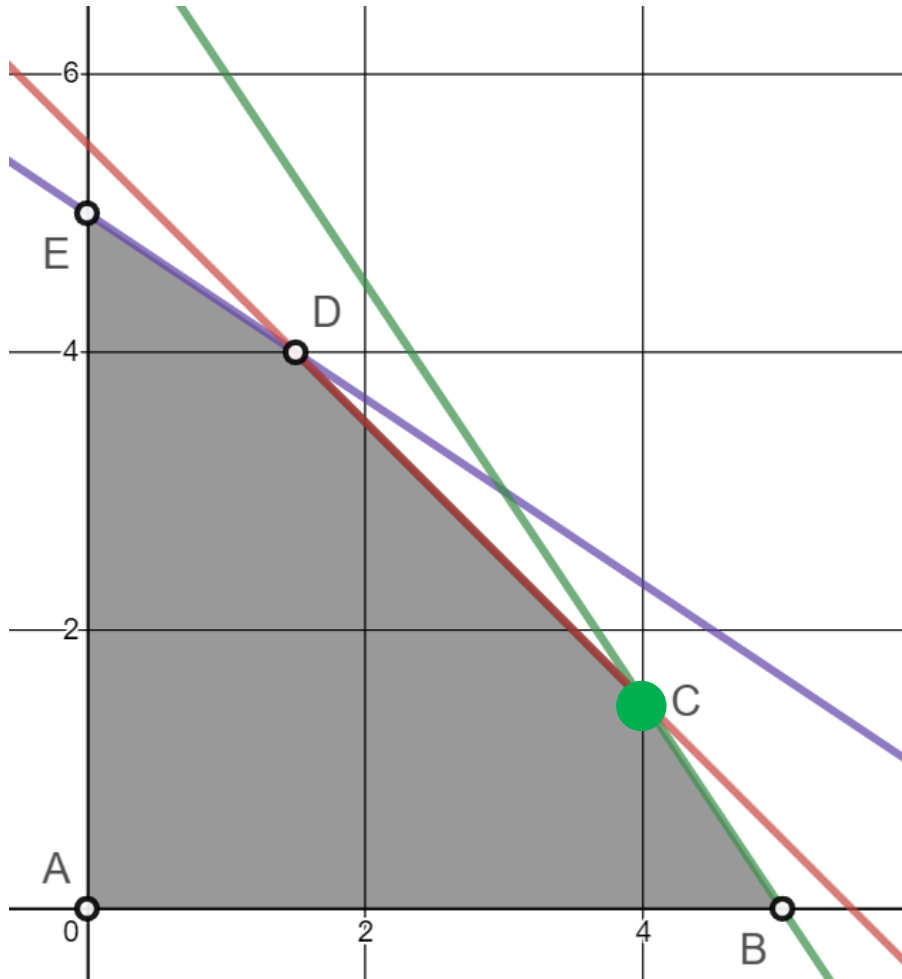
Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$
$C = 26$		



$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$
$C = 26$	$B = 25$	$D = 23.5$

Point D (1.5,4) has an objective value of
 $P = 5 \cdot 1.5 + 4 \cdot 4 = 23.5$



$$P = 5x + 4y$$

Current Point	Adjacent Points	
$A = 0$	$B = 25$	$E = 20$
$B = 25$	$A = 0$	$C = 26$
$C = 26$	$B = 25$	$D = 23.5$

Cannot improve, optimal solution at C!

Higher dimensions

